A Critical Analysis of Duolingo Chess Through the Cognitive Theory of Multimedia Learning and Social Constructivism

Aryn Davis (co-written with a classmate)
Technology, Media, and Learning, Columbia University
MSTU 4083

October 21, 2026

Duolingo is a popular learning platform that has made language education accessible to millions of users worldwide. In recent years, it launched three courses that deviate from its language courses: mathematics, music, and chess. Duolingo Chess, the newest addition, reimagines traditional chess learning through the accessible and engaging approach that characterizes its other courses.

This platform was chosen since we are interested in informal learning amongst adult learners. Interestingly, our team also has contrasting levels of chess experience, where one member had some prior experience playing chess, while the other had never taken a lesson before. The differing perspectives provided an opportunity to observe how it performs for intermediate and novice learners. Duolingo Chess therefore is a compelling case for us to evaluate whether an established, informal learning platform's designs effectively assist learners in attaining meaningful learning. This analysis critically examines the platform through two theoretical lenses: the Cognitive Theory of Multimedia Learning (CTML) and social constructivism, offering a holistic view into individual cognition and the social construction of knowledge. Through the analysis, we aim to identify successes of Duolingo Chess and explore opportunities to enhance effective learning.

Rationale for Choosing Theories

The first theoretical lens is CTML, which proposes that genuine learning occurs when learners process sensory information, organize them into mental representations, and connect them to prior knowledge. This theory contains principles that guide designs to better support these cognitive processes, revealing that a good presentation of content can optimize the construction of meaningful understanding. We chose CTML since it is relevant to Duolingo Chess's design approach; the learning platform is a prime example within education that prioritizes cognitive ease in its design, which allows the learning experience to be manageable and encouraging. Our analysis aims to recognize how its design aligns with CTML principles, while also critically examining their shortcomings.

The second lens for this analysis uses social constructivism, drawing on two of its concepts: Vygotsky's (1978) zone of proximal development, which emphasizes learning with support from a more knowledgeable other, and situated learning, which emphasizes learning through participation in a community's authentic practices (Brown et al., 1989; Barab & Duffy, 2012). We selected social constructivism to examine how an application meant for individual learning can teach a culturally rich game of chess, and where it may fall short. Chess has a deep history that includes contributions of different chess masters, key games that developed new tactics, and specific etiquette. Therefore, learning chess is not just about learning moves that lead to a win, but also about navigating the game's cultural aspects. Although Duolingo is designed for self-paced, independent learning, it includes social features, such as matches with friends and automated characters, which raise the question of whether learners make meaning together or simply practice side by side. Chess expertise shows in how easily a player recognizes and counters tactics, making potential opportunities for a more knowledgeable other to scaffold understanding of both the tactic and the opponent's intent.

CTML primarily focuses on the internal process. Thus, analysis through CTML offers an intrapersonal perspective and examines the relation between the learner and material. Social constructivism comes in as the perspective that centers on social and cultural contexts, where learning also occurs. Together, they offer a holistic critique that considers both cognition and socially mediated learning. However, these lenses can also pull in opposite directions: CTML's

principles may trim the cultural and historical material that social constructivism treats as essential, thus, our analysis weighs where that context deepens meaning and where it risks overloading novices.

Design Analysis

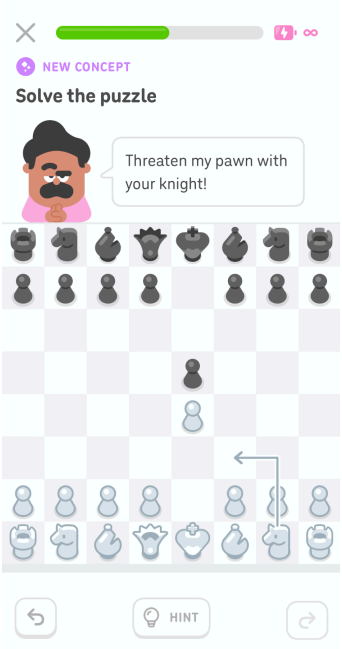
Cognitive Theory of Multimedia Learning

CTML is grounded in three cognitive assumptions that construct the architecture of human cognition: dual-channels, limited-capacity, and active processing. They form the foundation for CTML's three instructional design goals: reducing extraneous processing, managing essential processing, and fostering generative processing. Structured around these goals, the following analysis identifies Duolingo Chess's alignment and tensions with each.

Mayer's first instructional goal is to reduce extraneous processing. This means minimizing unnecessary cognitive effort expended on comprehending instructional material. Duolingo Chess executes this goal well. As portrayed in Figure 1, the explanatory text right above the chess board follows the spatial contiguity principle, which suggests that presenting relevant text and graphics in close proximity helps learners establish connections at ease. Duolingo Chess also maps arrows to indicate movement of a chess piece. Consistent with the signaling principle, these arrows reduce extraneous processing by directing attention to information that relates back to the text. Beyond the structural and visual cues, it's noticeable that the interface is minimalist. Through simplification, Duolingo Chess activates the coherence principle by eliminating unneeded information, such as excluding Algebraic Chess Notation, employing a reduced color palette, and presenting minimal text. This uncluttered environment further supports Mayer's first goal.

Figure 1

Duolingo Chess Lesson Interface



Note. The screenshot illustrates the arrangement of a lesson interface, which remains consistent throughout the lesson.

However, several tensions exist. First, the spatial contiguity principle is not fully demonstrated, since the text is not positioned beside the specific pieces. However, the current static placement is ideal, since learners need to consider the entire board; introducing floating text will instead obscure necessary information. Second, there is a misalignment to the redundancy principle, since graphics, audio narration, and its identical text are presented simultaneously. Despite this, the texts allow learners to take lessons even in places where audio is frowned upon; moreover, they include people that are hard of hearing. Although not aligned with CTML, this choice is essential since it forms an inclusive, accessible learning experience. Third, a tension with the coherence principle presents after completion of each lesson, where several screens of awards are displayed. These screens create distractions and unnecessary friction for learners that want to proceed to the next lesson.

Mayer's second instructional goal is managing essential processing, which refers to the cognitive effort required to construct mental representations. Duolingo Chess's design aligns with its segmentation principle, which suggests that presenting complex material in manageable chunks will help learners better digest material. Duolingo does this through dividing concepts into bite-sized lessons. Even within each lesson, the principle is demonstrated, where learners press "Continue" after completing each set of moves. This approach allows users to digest concepts before moving on to new material, helping manage essential processing.

However, we noticed a notable misalignment: Duolingo Chess does not implement the pre-training principle, which suggests that learners process material better when first introduced to rudimentary information of the concepts. When entering a lesson on Duolingo Chess, learners are immediately introduced to moves without knowing what makes them strategically useful. For example, when entering Section 2 Unit 6 titled "The Italian Game", there is no explanation of what an opening is, and why it is important. Learners are often in a liminal space between executing moves and knowing their purpose. The insufficient preparation increases cognitive effort, since learners have to somehow organize the meaning-detached moves into mental representations.

Mayer's third instructional goal is to foster generative processing. This goal utilizes social cues that trigger learners' motivation to put in mental effort. Duolingo Chess executes this well with Oscar, the character that acts as the guide. Figure 1 captures Oscar saying, "Threaten my pawn with your knight". This reflects the personalization principle, where Oscar utilizes conversational language like "my" and "your", tightening the social proximity between the guide and learner. Extending Mayer's CTML, Mayer and DaPra (2012) found that integrating animated guides improved performance in transfer tasks. This is relevant to chess, a transfer-based subject. Oscar epitomizes this finding; instead of merely exhibiting expressions on static characters, it incorporates gestures that create a more life-like representation. This design choice encourages learners to invest cognitively and achieve genuine learning.

Fiorella and Mayer (2015) identify learning strategies to foster generative processing beyond social cues. One strategy is self-testing, where learners are prompted to retrieve information from long-term memory. Duolingo Chess does not employ these strategies. Although incorporation of unguided problems are present, they only assess the ability of executing moves, not the purpose behind them. Learners may complete lessons through trial-and-error, without exercising mental effort to connect to prior knowledge.

Social Constructivism

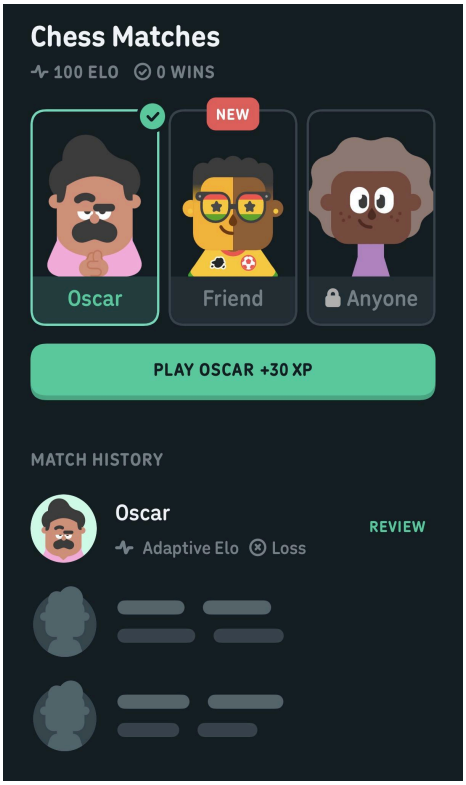
Social constructivism maintains that knowledge is constructed through meaningful engagement in a community setting, often with support from someone with more knowledge(Vygotsky, 1989). Chess is a two-player game, where knowledge of tactics and how to best counter them comes from practice. The focus of this analysis will be on the two areas that support both a social constructivism and situated learning approach: scaffolding and social opportunities to engage in the culture.

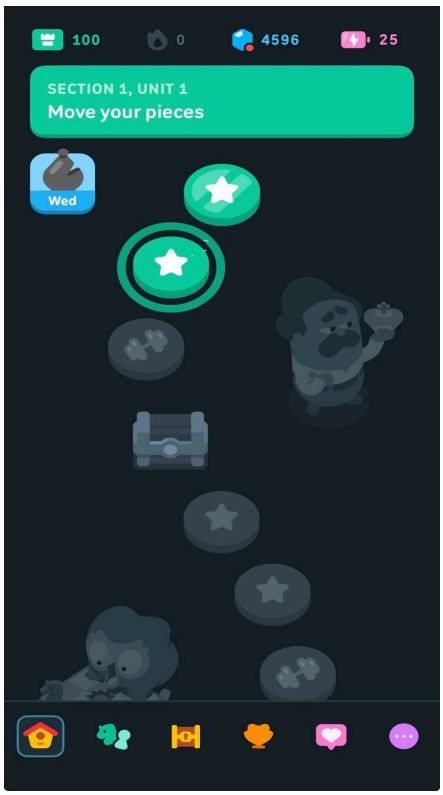
The first critical process needed for social constructivist learning is proper scaffolding. This idea of scaffolding pairs nicely with supporting the zone of proximal development as cited in Vygotsky's work. Students should be able to recognize where their skills currently lie and use the instruction from Duolingo to slowly expand their skill level. There is active scaffolding available after users select their mastery level at the beginning of the course. Users who have some familiarity with chess can jump into harder lessons that detail more advanced tactics, while novices can start from learning the moves of each piece. Our experience with Duolingo chess highlighted some of the scaffolds available to help users build these tactics without overwhelming them or forcing too much memorization, but it often failed to reinforce practice to help users master them. We must assume that the Duolingo automated characters function as the knowledgeable other in this context, as they lead us through the lessons. While the app may suggest a learner return to review concepts after a while or when they seem to be struggling, this usually happens after multiple failed attempts across multiple lessons and feels more like a review than a chance to achieve mastery.

This scaffold does not extend to the games tab. Duolingo Chess allows users to play chess matches with an automated character, a friend, or a random Duolingo user. The automated character matches the learner's skill level in this games tab, but the platform offers no hints to help learners understand which moves are best. Instead of a gradually fading support system highlighted as a positive approach from Brown et al. (1989), the app removes all support outside of the lesson context. This proves difficult for a total beginner who is unsure of gameplay to develop meaningful experience and become familiar with level-appropriate strategies for their match. Matches with friends or random users also don't offer scaffolded support, so even in a social context, the learner must figure out the game on their own.

Figures 2 and 3

The games tab for Duolingo Chess, where users can play a match for practice, and the lesson tab for novice learners, where basic mastery of one lesson allows progression to the next.





The second property needed for social constructivist analysis is how Duolingo Chess uses and introduces users to chess's cultural knowledge. One thing Duolingo does well is maintain the concept of an Elo rating for all chess learners. An Elo rating is a number used across chess tournaments and online platforms alike, providing a standard system for determining relative ranking and mastery. By using this rating system, Duolingo Chess affirms its commitment not only to teaching learners chess, but also to placing them in the sociocultural context of chess worldwide. Users can take their ratings to other platforms and play comparative games with others of the same or slightly higher rating, preventing an experience that is past the learner's zone of proximal development and potentially discouraging them from further study. However, Duolingo Chess lacks other etiquette, such as knowing when your position is too far gone to recover and should result in resignation, and understanding the time clock for in-person games. Players can learn these norms by participating in the chess community, so not incorporating them here reduces Duolingo Chess's ability to serve as a community of practice (Barab & Duffy, 2012). However, Duolingo should not incorporate too much of this context, as it could violate CTML's coherence principle and potentially overload novices.

Figure 4

The Elo ratings chart used across chess.



What do different Elo ratings mean?

Rating range	Level
Under 800	Beginner
800–1200	Casual player
1200–1500	Intermediate
1500–1800	Club player
1800–2200	Advanced / Expert
2200+	Master and above

Redesign Recommendations

Cognitive Theory of Multimedia Learning

The analysis through CTML reveals a few opportunities for improvement. First, reducing post-lesson reward screens; second, introducing foundational concepts at the beginning of each lesson; and third, integrating reflective quizzes within each lesson.

The first improvement concerns the post-lesson reward screens. The multiple screens displaying various achievements become interruptions to those that seek to complete multiple lessons. We recognize that these screens serve as motivators for user retention. Thus, rather than removing them, we propose merging achievements into one screen, which will present a stronger alignment to reduce extraneous processing. For the second improvement, we propose adding basic information at the start of each lesson. In order to align with CTML, we propose to deliver essential information through minimal text, ensuring that the learning experience remains cognitively undemanding. For example, at the beginning of introducing “The Italian Game,” a short message saying, “A chess opening is the first phase of a game, where players develop their pieces strategically. You will learn the Italian Game first.” Then, followed by “The Italian Game brings out your bishop to pressure the king.” By integrating pre-training, essential processing will be assisted. The third improvement is integrating quizzes for self-testing. Currently, learners are merely taught to reproduce moves without comprehending purpose. Integration of reflective, free-text quizzes that target semantics of chess encourages retrieval from long-term memory, aligning with the third goal.

Social Constructivism

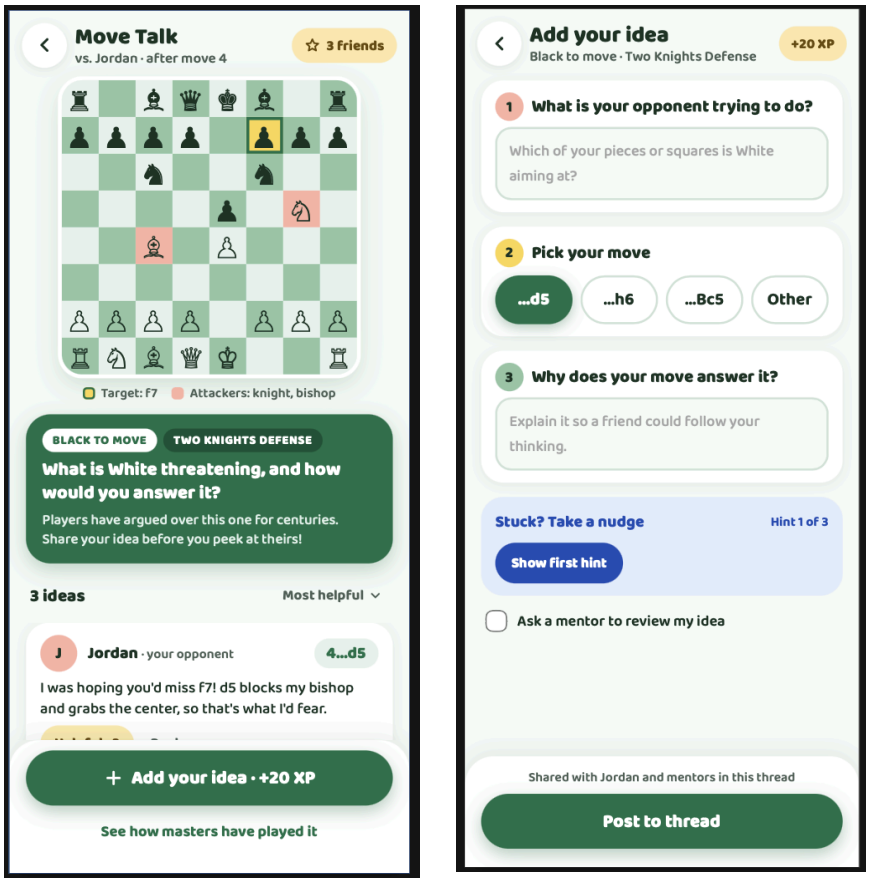
In order to align Duolingo chess more with a social constructivist lens, we propose a few changes, including historical knowledge and etiquette and a collaborative space for discussion during matches. As in the CTML-aligned redesign, providing context for what a tactic is creates more meaningful learning and reflects an authentic integration of chess culture (Brown et al., 1989). This can also extend to the games tab, where a brief prompt can let a learner know when they have no further moves and must resign.

Another improvement is adding a “Move Talk” space where players can discuss potential moves, anticipate opponents' outcomes, and have their logic reviewed by mentors or friends, introducing a knowledgeable other that isn't automated. This is missing from the games tab and

would support social negotiation of meaning in learning. An optional hints option that becomes less frequent as the game progresses would also add a gradually fading support system.

Figure 5, 6

Mockups of Duolingo chess match interface that incorporates social interaction and scaffolding for collective meaning making



Conclusion

With the critical analysis of Duolingo Chess through CTML and social constructivism, we have identified the platform's strengths and weaknesses in its instructional design. Through CTML, we found that the simplicity in lesson interface and brevity in lesson content aligns well with unloading extraneous cognitive effort. However, major gaps remain for promoting better cognitive representation and effort. From a social constructivism perspective, the emphasis on accessibility limits learners' ability to construct knowledge through cultural and social contexts. Although Duolingo Chess demonstrates the value of accessible instructional design, it could better balance simplicity with genuine cognitive engagement and socially situated learning.

References

Barab, S. A., & Duffy, T. (2012). From practice fields to communities of practice. In *Theoretical foundations of learning environment*s (pp. 29-65). Routledge.

Brown, J. S., Collins, A., & Duguid, P. (1989). Situated cognition and the culture of learning. *Educational Researcher, 18*(1), 32–42.

Fiorella, L., & Mayer, R. E. (2015). Learning as a generative activity: Eight learning strategies that promote understanding. Cambridge University Press. https://doi.org/10.1017/CBO9781107707085

Mayer, R. E. (2017). Using multimedia for e-learning. Journal of Computer Assisted Learning, 33(5), 403–423. https://doi.org/10.1111/jcal.12197

Mayer, R. E., & DaPra, C. S. (2012). An embodiment effect in computer-based learning with animated pedagogical agents. Journal of Experimental Psychology: Applied, 18(3), 239–252. https://doi.org/10.1037/a0028616

Vygotsky, L. S. (1978). *Mind in society: The development of higher psychological processes* (M. Cole, V. John-Steiner, S. Scribner, & E. Souberman, Eds.). Harvard University Press.